

Background

High-intensity focused ultrasound (HIFU) is an emerging non-invasive modality for thermal ablation of soft-tissue tumours. Clinical deployment is bottlenecked by two computationally expensive sub-problems: *forward* simulation of the acoustic pressure field in patient-specific tissue, and *inverse* synthesis of the transducer phase vector that steers energy to a desired focal point.

Full-wave solvers such as k-Wave deliver reference accuracy but are too slow for iterative planning, while direct inverse regression of phases is hampered by a one-to-many target–phase mapping with an inherent gauge degree of freedom. We present a two-track neural surrogate that addresses both sub-problems on anatomically realistic breast phantoms drawn from the recently released **OpenBreastUS** dataset (Zeng *et al.* 2025).

Track A — Forward surrogate

One thousand 316×256 sound-speed maps were simulated in 2-D time-domain k-Wave with a 12 mm water stand-off and a 1 MHz focused transducer. A 2-D Fourier Neural Operator (FNO) and a matched U-Net baseline were trained with a combined $L_p\text{Loss} + 0.3 \cdot H_1\text{Loss}$ objective on an identical 70 / 15 / 15 split.

FNO achieved a test relative L2 error of **0.097**, outperforming the U-Net baseline (**0.264**) by **2.7×**. Qualitative inspection further shows that FNO recovers the full refraction tail that the U-Net smears, consistent with FFT-based representations matching the spectral structure of wave propagation. Validation-loss scaling from 0.81 (200 samples) to 0.39 (1 000 samples) indicates substantial further headroom on the full 8 000-sample dataset. A backbone-robustness ablation (FNO vs U-Net vs ConvNeXt-2D, 0.99 $L_p\text{Loss}$) confirms that the spectral prior is the dominant effect — generic CNN backbones do not transfer.

Track B — Inverse planning

A diagnostic study showed that the heat-deposition volume Q and the transducer phase vector are not related by a single-valued map: the circular-cosine similarity of nearest-neighbour phase sets in target space is **0.002** $\pm$ **0.043**, indistinguishable from noise. This one-to-many pathology is exacerbated by a continuous gauge symmetry (identical phases modulo a global constant).

We verified the gauge analytically, numerically on synthetic 256-element arrays, and — independently — through full-wave k-Wave simulations by our ITU collaborator. A constant $+20^\circ$ phase offset produced a measured focal displacement of **0 mm** and a sub-1 % acoustic-intensity change in all three verifications. We therefore recast the inverse task as **3-DOF focus-point regression**: a compact 3-D CNN (0.89 M parameters, trained on only 30 simulations) predicts the focal coordinate from Q , and the transducer phases are subsequently obtained analytically by delay-and-sum beamforming.

On an unseen test split the network attains lateral RMS accuracies of **5.1 mm (X)** and **5.3 mm (Y)**, well within the native HIFU focal-spot diameter, with axial accuracy of 26.9 mm dominated by the elongated focal zone and by the Z-range being twice the lateral range. The same 3-D CNN outperforms the strongest classical heat-map localiser (amplitude-weighted centroid) by **2.5×** on lateral error.

Architecture ablations

In a multi-seed ablation at matched ~ 0.9 M parameter budgets (three seeds, 120 epochs per run), the plain CNN achieves 4.63 ± 2.21 mm (X) and 2.69 ± 1.21 mm (Y) lateral RMS. A ResNet-3D backbone and a multi-scale U-Net encoder give statistically equivalent lateral numbers (X 5.9 ± 3.8 / 3.9 ± 0.7 mm; Y 3.1 ± 1.4 / 3.0 ± 1.1 mm) — all well inside the native focal spot. The CNN retains the lowest combined RMS (26.25 ± 4.61 mm vs 34.48 / 34.17 mm) thanks to stronger axial behaviour. The consistency of lateral accuracy across three distinct backbones indicates that the achievable precision is **a property of the problem structure, not of any particular architecture**.

We additionally evaluated a state-of-the-art 3-D U-Net heatmap-regression variant (nnLandmark / H3DE-Net style, 2025). Trained with a differentiable soft-argmax (DSNT) loss — after pure-MSE supervision collapsed on our 30-sample regime — the plain heatmap variant reaches parity with the direct-coordinate CNN (25.27 vs 26.25 mm overall test RMS) and is slightly better on the axial axis (Z: 23.79 vs 25.65 mm). Adding a per-voxel sub-voxel offset head (H3DE-Net style) triples the error, as 22 training samples cannot constrain the dense offset regression. This ablation empirically **defines the sample-size threshold** at which the heatmap-regression advantage reported in the nnLandmark family (500–1000+ samples, 10–30 % improvement) begins to emerge.

Phase-quantisation study

A phase-quantisation study (5° , 10° , 15° step) showed that the focus is stable under coarse phase rounding — less than **0.35 %** acoustic- intensity error at a 5° step and sub-0.1 mm focal displacement — enabling a discrete, gauge-fixed output space for future phase regressors.

Significance

To our knowledge this is the first application of OpenBreastUS to HIFU treatment planning (the original release targets ultrasound computed tomography reconstruction), and the first demonstration of an end-to-end AI-assisted pipeline — FNO forward surrogate + learned focus localisation + analytical beamforming — that bypasses both the ill-posedness of direct phase regression and the computational cost of conventional full-wave planning.

The work contributes (i) a spectrally-consistent neural operator baseline for 2-D breast-tissue HIFU, (ii) an empirical gauge-and-quantisation characterisation of the inverse problem that is of independent value for classical phase optimisers, and (iii) a sample-efficient focus-point network that becomes a building block for real-time, patient-specific treatment planning.

Keywords: HIFU, neural operator, Fourier Neural Operator, k-Wave, OpenBreastUS, focal beamforming, gauge symmetry, tumour ablation.